E-classroom	**Name: Memorandum**					
	Gr 10		**Date:**		**Time:**	**2HR**
CAPS Reference	**Functions**					
Topic	**Parabolic Function**					

2.1 $\boldsymbol{f(x) = -x^2 + 4}$

$\underline{y - intercept; x = 0}$

$y = -(0)^2 + 4$

$= 4$

$(0; 4)$

$\underline{x - intercept; y = 0}$

$0 = -x^2 + 4$

$0 = -(x^2 - 4)$

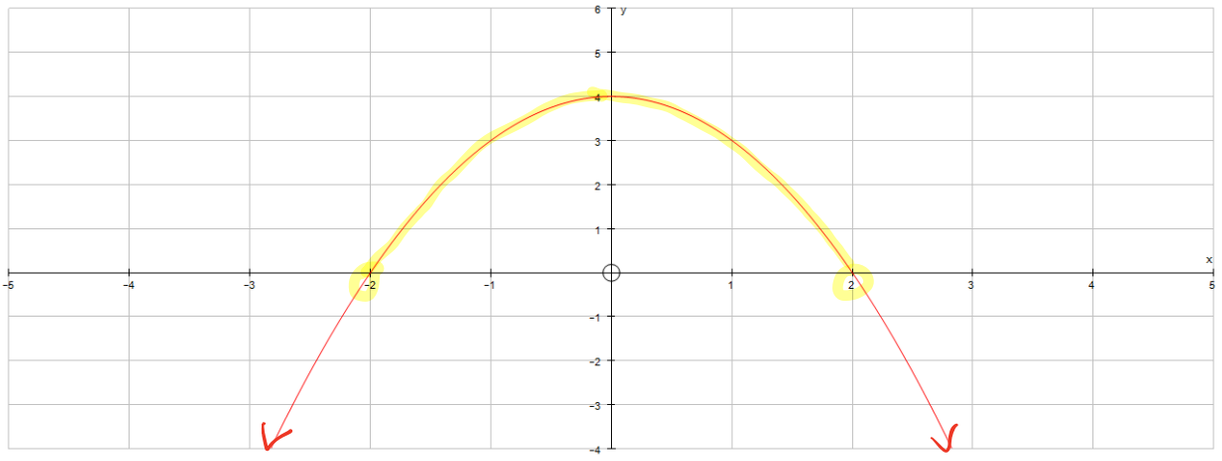
$0 = -(x + 2)(x - 2)$

$0 = (x + 2)(x - 2)$

$x + 2 = 0 \quad or \quad x - 2 = 0$

$x = -2 \quad or \quad x = 2$

$(-2; 0) \quad or \quad (2; 0)$



a) $y \in$ ~~$(4; \infty)$~~ $(-\infty ; 4]$

b) $(0; 4)$

c) ~~$x < -2 \text{ or } x > 2$~~; $x \in R$ $x \in (-2 ; 2)$

d) $h(x) = f(x) + 2$

$= -x^2 + 4 + \mathbf{2}$

$= -x^2 + 6$

2.2 $\boldsymbol{g(x) = -3x^2 + 27}$

$\underline{y - intercept; x = 0}$

$y = -3(0)^2 + 27$

$= 27$

$(0; 27)$

$\underline{x - intercept; y = 0}$

$0 = -3x^2 + 27$

$0 = -3(x^2 - 9)$

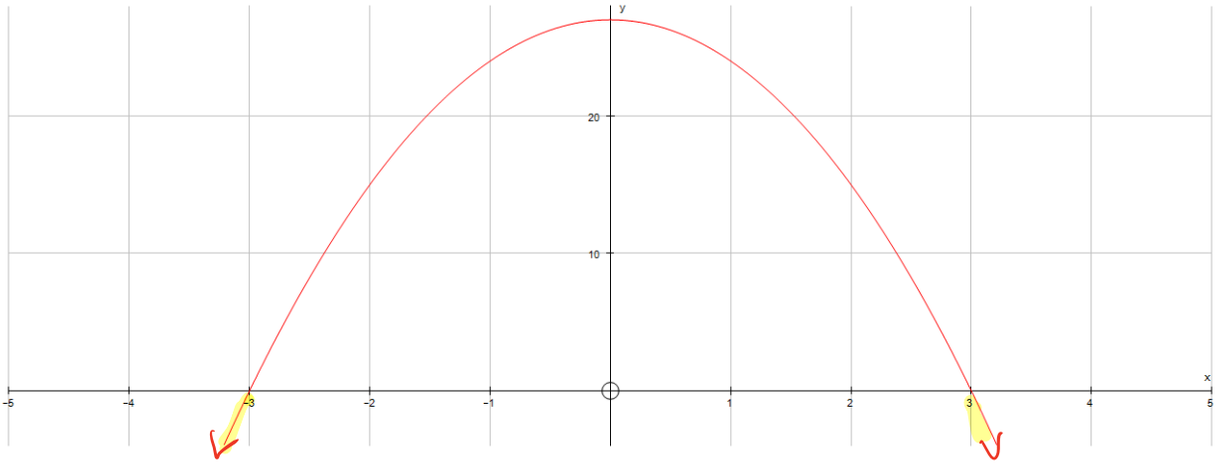
$0 = -3(x + 3)(x - 3)$

$0 = (x + 3)(x - 3)$

$x + 3 = 0 \quad or \quad x - 3 = 0$

$x = -3 \quad or \quad x = 3$

$(-3; 0) \quad or \quad (3; 0)$



a) $y \in$ ~~$(27; -\infty)$~~ $(-\infty; 27]$
b) $(0; 27)$
c) $x = 0$
d) $x \leq -3 \; or \; x \geq 3; x \in R$
e) $h(x) = -g(x) - 2$

$= -(3x^2 + 27) - \mathbf{2}$

$= -3x^2 - 27 - 2$

$= -3x^2 - 29$

2.3 $\boldsymbol{f(x) = \frac{1}{2}x^2 - 1}$

$\underline{y - intercept; x = 0}$

$y = \frac{1}{2}(0)^2 - 1$

$= -1$

$(0; -1)$

$\underline{x - intercept; y = 0}$

$0 = \frac{1}{2}x^2 - 1$

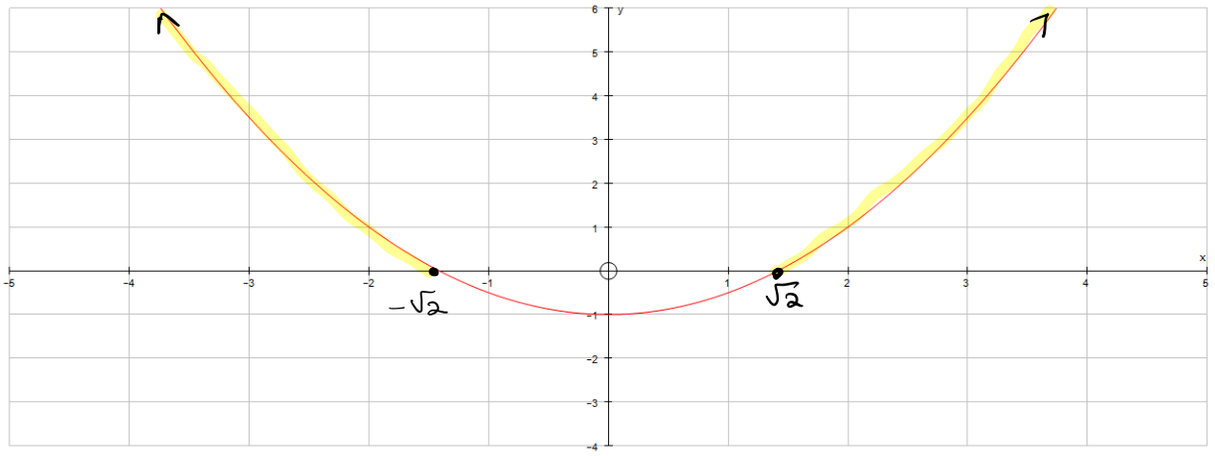
$\times 2) \quad 0 = x^2 - 2$

$2 = x^2$

$\mp\sqrt{2} = x$

$x = -\sqrt{2} \ \ or \ \ x = \sqrt{2}$

$(-\sqrt{2}; 0) \ \ or \ \ (\sqrt{2}; 0)$



a) $y \in [-1; \infty)$
b) $(0; -1)$
c) $x \leq -\sqrt{2} \ or \ x \geq +\sqrt{2}; x \in R$

d) $h(x) = f(x) + 3$

$= \frac{1}{2}x^2 - 1 + \mathbf{3}$

$= \frac{1}{2}x^2 + 2$

$\therefore \boldsymbol{Range: y} \in [\mathbf{2}; \infty)$

$$f(x) = \frac{1}{2}x^2 - 1$$

-1: Is the minimum value of the function and the starting point of the range.

Therefore you could calculate the new range as such $-1 + 3 = 2$

2.4 $\boldsymbol{g(x) = -\frac{1}{3}x^2 + 2}$

$\underline{y - intercept; x = 0}$

$y = -\frac{1}{3}(0)^2 + 2$

$= 2$

$(0; 2)$

$\underline{x - intercept; y = 0}$

$0 = -\frac{1}{3}x^2 + 2$

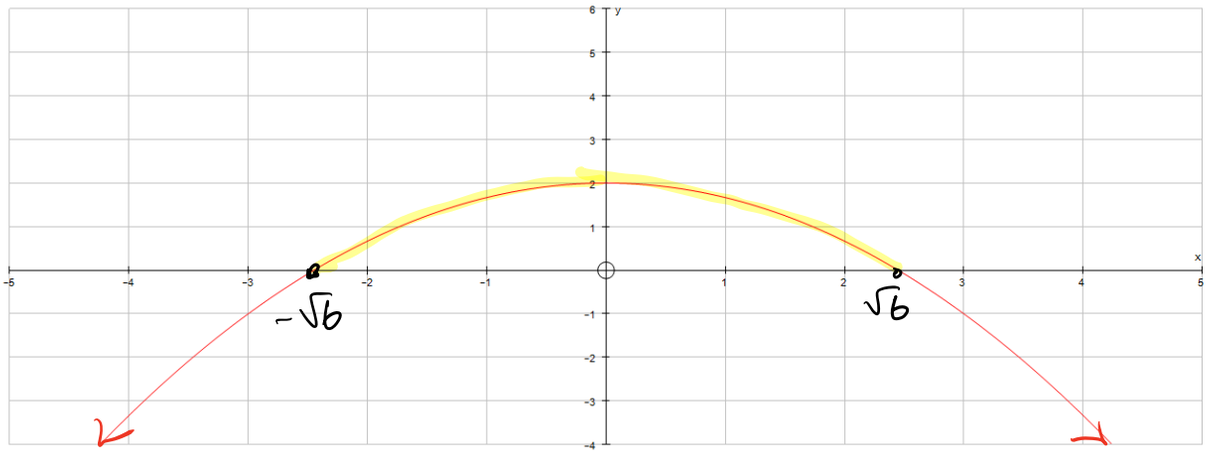
$\times -3) \quad 0 = x^2 - 6$

$6 = x^2$

$\mp\sqrt{6} = x$

$x = -\sqrt{6} \ \ or \ \ x = 6$

$(-\sqrt{6}; 0) \ \ or \ \ (\sqrt{6}; 0)$



a) $y \in$ ~~$(2; -\infty)$~~ $(-\infty ; 2]$
b) $(0; 2)$
c) $-\sqrt{6} < x < \sqrt{6}; x \in R$
d) $x = 0$
e) $h(x) = -g(x) + 1$

$= -\left(-\frac{1}{3}x^2 - 2\right) + 1$

$= \frac{1}{3}x^2 - 2 + 1$

$= \frac{1}{3}x^2 - 1$

$\therefore \boldsymbol{Range}: \boldsymbol{y} \in [-\mathbf{1}; \infty)$

~~$= -\left(-\frac{1}{3}x^2\right)$~~

$f(x) = -\frac{1}{3}x^2 + 2$

2: Is the maximum value of the function and the starting point of the range.

Therefore you could calculate the new range as such $-(2) + 1 = -1$

4.1

Step 1: Substitute the turning point $(0; 2)$

$y = ax^2 + q$

$y = ax^2 + 2$

Step 2: Substitute the given coordinate $(1; 3)$

$y = ax^2 + 2$

$3 = a(1)^2 + 2$

$3 = a + 2$

$a = 3 - 2$

$a = 1$

Step 3: Substitute a in to the equation.

$$\boldsymbol{f(x) = x^2 + 2}$$

4.2

Step 1: Substitute the turning point $(0; 5)$

$y = ax^2 + q$

$y = ax^2 + 5$

Step 2: Substitute the given coordinate $(-1; 8)$

$y = ax^2 + 5$

$8 = a(-1)^2 + 5$

$8 = 1a + 5$

$a = 8 - 5$

$a = 3$

Step 3: Substitute a in to the equation.

$$\boldsymbol{f(x) = 3x^2 + 5}$$

4.3

Step 1: Substitute the turning point $(0; 10)$

$y = ax^2 + q$

$y = ax^2 + 10$

Step 2: Substitute the given coordinate $(2; 6)$

$y = ax^2 + 10$

$6 = a(2)^2 + 10$

$6 = 4a + 10$

$4a = 6 - 10$

$4a = -4$

$a = \frac{-4}{4}$

$a = -1$

Step 3: Substitute a in to the equation.

$$\boldsymbol{f(x) = -x^2 + 10}$$

4.4

Step 1: Substitute the $x - intercepts(-2; 0)\ and\ (2; 0)$

$y = a(x - x_1)(x - x_2)$

$y = a(x - (-2))(x - 2)$

$y = a(x + 2)(x - 2)$

Step 2: Substitute the given coordinate $(1; -6)$

$-6 = a(1 + 2)(1 - 2)$

$-6 = a(3)(-1)$

$-6 = -3a$

$\frac{-6}{-3} = a$

$2 = a$

Step 3: Substitute a in to the equation.

$y = a(x + 2)(x - 2)$

$y = 2(x + 2)(x - 2)$

$y = 2(x^2 - 4)$

$y = 2x^2 - 8$

$$\boldsymbol{f(x) = 2x^2 - 8}$$

4.5

Step 1: Substitute the $x - intercepts(-6; 0)\ and\ (6; 0)$

$y = a(x - x_1)(x - x_2)$

$y = a(x - (-6))(x - 6)$

$y = a(x + 6)(x - 6)$

Step 2: Substitute the given coordinate $(-4; -20)$

$-20 = a(-4 + 6)(-4 - 6)$

$-20 = a(2)(-20)$

$-20 = -20a$

$\frac{-20}{-20} = a$

$1 = a$

Step 3: Substitute a in to the equation.

$y = a(x + 6)(x - 6)$

$y = 1(x + 6)(x - 6)$

$y = 1(x^2 - 36)$

$y = x^2 - 36$

$$\boldsymbol{f(x) = x^2 - 36}$$